

# THE INDEPENDENCE ASSUMPTION FOR ROUNDING ERRORS IN BLOCK FLOATING POINT: FAILURE OF CONDITIONAL SYMMETRY AND ITS EFFECT ON SUMMATION ERROR ACCUMULATION\*

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**Abstract.** Error analyses of block floating point (BFP) and related shared-exponent formats, including the MXFP family used by modern tensor cores, usually assume that within-block quantization errors are zero-mean and mutually uncorrelated. That assumption dates to a 1996 analysis of BFP roundoff and is rarely re-tested on modern data. We re-examine it by Monte Carlo simulation and reduce the outcome to a parameter-free model.

Four rigorous facts carry the study. Without extra assumptions,  $\text{Cov}(e_i, e_j) = \mathbb{E}[\text{Cov}(e_i, e_j \mid \xi)] + \text{Var}(m(\xi))$  with  $m(\xi) = \mathbb{E}[e \mid \xi]$ , so any failure of the classical model is a block-level effect. The variance inflation factor satisfies  $0 \leq \mathcal{I}_n \leq n$ . The variance law  $\mathbb{E}[e^2] = \mathbb{E}[s^2 \gamma(\xi)]$  makes the classical model exactly the case  $\gamma \equiv 1/12$ ,  $\delta \equiv 0$ , and separates the two defects:  $\delta \not\equiv 0$  creates correlation,  $\gamma \not\equiv 1/12$  biases the variance formula. Under unbounded support with  $\mathbb{E}[x_1^2] < \infty$ ,  $m(\xi) \rightarrow -\mathbb{E}[x_1]$ ,  $\text{Var}(m(\xi)) \rightarrow 0$  and  $\mathcal{I}_n/n \rightarrow 0$ , so linear accumulation is transient.

The reduced model writes  $m(\xi) = s \delta(s)$  and  $\mathbb{E}[e^2 \mid \xi] = s^2 \gamma(s)$ : one-dimensional response functions times the block maximum, with no adjustable parameter. Over 36 round-to-nearest configurations it matches measured  $\mathcal{I}_n$  within 4.5% (median 1.15%). Simulations cover six distributions and four mantissa widths. Zero-mean fails for asymmetric inputs (lognormal  $z = -126.8$ , ReLU  $z = -39.5$ ) and holds for symmetric ones. For ReLU the measured variance-to-classical ratio is 0.506, matching the atom prediction  $1 - p_0 = 0.500$ ; under stochastic rounding the same argument predicts 1.000 against a measured 1.015. A decomposition given  $\xi$  shows that within-block correlation comes almost entirely from  $\text{Var}(m(\xi))$ . Bounded support forces  $\mathcal{I}_n \rightarrow 1$ , whereas unbounded support gives an inflation that first rises and then falls; under stochastic rounding all 24 configurations give  $\mathcal{I}_n \approx 1$ .

**Key words.** block floating point, low-precision arithmetic, rounding error analysis, probabilistic error bounds, stochastic rounding, tensor cores

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**1. Introduction.** Low-precision arithmetic now dominates many workloads on modern accelerators. NVIDIA tensor cores, AMD matrix cores, and the OCP microscaling (MX) formats raise throughput by sharing a single exponent among several elements. Formats of this kind are called block floating point (BFP): the  $n$  values of a block share one exponent  $\xi$ , and each element stores only a mantissa. Relative to elementwise floating point this saves exponent bits but reduces the dynamic range inside the block.

For decades, error analyses of such formats have relied on a statistical model inherited from Kalliojärvi and Astola [21]: quantization errors are zero-mean, their variance is determined by the block exponent and the mantissa width, and, in later citations, the errors may also be treated as uncorrelated [22, 25, 12, 17]. In parallel, numerical analysis developed probabilistic rounding error bounds: under zero-mean and (approximately) independent errors, summation and prefix errors grow like  $O(\sqrt{n}u)$  rather than  $O(nu)$  [18, 20, 16], and the framework has been extended to mixed precision and stochastic rounding [19, 10, 3, 4, 8, 9, 11], to multiword tensor-core arithmetic [13], and to integer-splitting emulation [1].

The two communities have not fully met: hardware-oriented papers often assume the statistical model without re-validating it on modern formats and data, while numerical analysis papers test such models on elementwise floating point, multiword arithmetic, integer slicing, or stochastic rounding, none of which contains a data-determined shared exponent. This paper studies that missing case directly.

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## Contributions.

1. Covariance decomposition (Proposition 2): without further assumptions,  $\text{Cov}(e_i, e_j) = \mathbb{E}[\text{Cov}(e_i, e_j \mid \xi)] + \text{Var}(m(\xi))$ , so any failure of the classical model is a block-level effect. Conditional exchangeability, usually stated as an extra hypothesis, is automatic here (Lemma 1) and yields  $0 \leq \mathcal{I}_n \leq n$  (Corollary 3).
2. Exact variance laws and a reduced model (Proposition 7, Lemma 8):  $\mathbb{E}[e^2] = \mathbb{E}[s^2 \gamma(\xi)]$  and  $m(\xi) = s \delta(s)$  with  $\delta, \gamma$  depending only on the distribution and the step, so all remaining randomness is the block maximum. There is no adjustable parameter.
3. Exact stochastic-rounding counterpart (Proposition 5): the conditional covariance and  $\text{Var}(m(\xi))$  both vanish, so  $\mathcal{I}_n \equiv 1$  and  $\text{Var}(e) = \mathbb{E}[s^2 \theta(1 - \theta)]$  (Corollary 6). Stochastic rounding is therefore a falsifiable probe of the mechanism, and not merely a better rounding rule.
4. Monte Carlo experiments over six distributions and four mantissa widths: failure of zero-mean for asymmetric inputs (subsection 5.1); structural bias of the classical variance formula with its atom explanation (subsection 5.2); rise and subsequent fall of the inflation with block length (subsection 5.3); a direct decomposition given  $\xi$  together with a numerical test of (A2) (subsection 5.6); and a parameter-free model that stays within 4.5% of measured  $\mathcal{I}_n$  over 36 configurations (subsection 5.7).
5. Two support regimes (subsections 3.5 and 3.6): bounded support gives  $\mathcal{I}_n \rightarrow 1$  exponentially fast and independently of the endpoint tail index (Proposition 10), while unbounded support with  $\mathbb{E}[x_1^2] < \infty$  gives  $\rho_n \rightarrow 0$  and  $\mathcal{I}_n/n \rightarrow 0$  (Proposition 12). No admissible distribution in this setting yields asymptotic linear accumulation  $\mathcal{I}_n \asymp n$ .

**Relation to concurrent work.** Zhang et al. [27] already observed, in the fault-tolerance context of a BFP-based NPU, that the shared exponent couples the elements of a block; our contribution is to quantify that coupling as a pairwise covariance identity with separately falsifiable predictions (Propositions 2 to 7, Propositions 10 and 12). Ang et al. [2] obtain  $\mathbb{E}[(S - \hat{S})^2] = k \mathbb{E}[D^2] + k(k-1)(\mathbb{E}[D])^2$  for scalar quantization in matrix multiplication, where the quadratic term can be “designed away” by a companding quantizer; a shared exponent is the complementary case, because  $\text{Var}(m(\xi))$  persists as  $n$  grows, so raising the quantization precision cannot remove the quadratic term. Sao et al. [23] obtain a structurally identical centred/non-centred dichotomy for reduction trees, but condition on the state of the tree and assume conditionally unbiased rounding. The grid biases of Zhao et al. [28] and Wu et al. [26] are defined in *amplitude space* and stay sign-symmetric for sign-symmetric inputs, so they do not amount to  $\mathbb{E}[e_i] \neq 0$ ; our claim is that the expectation of the signed error fails when the *data* are conditionally asymmetric (subsection 5.1). Finally, Proposition 5 is cognate with dither theory: Schuchman [24] and Gray–Stockham [15] long ago showed that suitable dithering makes the quantization error independent of the input; what we add is the use of that fact—driving the unobservable  $m(\xi)$  to zero—as a falsifiable probe.

This paper is predominantly a numerical study: theorem-level results are confined to Propositions 2 to 7, Lemma 8, Corollaries 3 to 6, and Propositions 10 and 12; all remaining statements about scaling behaviour are explicitly labelled as numerical observations or as unproved.

## 2. Background: an inherited and never re-validated assumption.

**2.1. Block floating point formats.** Fix a block length  $n$  and a mantissa width  $L_m$ . The shared exponent of the real values  $x_1, \dots, x_n$  in a block is taken to be

$$(1) \quad \xi = \left\lfloor \log_2 \max_{1 \leq i \leq n} |x_i| \right\rfloor, \quad s = 2^{\xi - L_m},$$

TABLE 1

The propagation chain of the classical assumption. For each reference the table states the property inherited from the 1996 result and how.

| Reference                           | Inherited property                                                                                                                                                         |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Song, Liu, and Wang, AAAI 2018 [25] | take the quantization error of a block to have zero mean and variance $\sigma^2$ , attributing both directly to Kalliojärvi and Astola (1996)                              |
| Lian et al., IEEE TVLSI 2019 [22]   | model $e_{bm}$ , the quantization error of the block mantissa, as an uncorrelated random variable rather than deriving zero-meanness for a data-determined shared exponent |
| Han et al., DAC 2025 [17]           | state that for a block floating point using round to nearest the quantisation error is zero-mean with variance $\sigma^2$ , citing their [31], which is the 1996 reference |

where  $s$  is called the *step* of the block. The quantized value of the  $i$ th element is  $\hat{x}_i = \text{RN}(x_i/s) \cdot s$ , where RN denotes rounding to nearest, ties to even. The error is defined as

$$(2) \quad e_i = \hat{x}_i - x_i, \quad i = 1, \dots, n.$$

Equations (1) and (2) are an idealized model of the OCP MX format family and of classical BFP: a shared exponent, rounding to nearest, no overflow or underflow. Real hardware may use other rounding rules or sticky-bit handling; subsection 5.4 analyses the sensitivity to the rounding rule, and a comparison against real hardware is left to future work (subsection 6.2).

**2.2. The inherited assumption and its propagation chain.** In a deterministic signal-processing setting, Kalliojärvi and Astola [21] established two properties of the BFP quantization error, which we call the *classical model*:

$$(3) \quad (\text{H1, zero mean}) \quad \mathbb{E}[e_i] = 0, \quad (\text{H2, variance}) \quad \text{Var}[e_i] = \mathbb{E}[s^2]/12.$$

Later work inherited (H1) and (H2) when citing that result, and also introduced a third, stronger and much less discussed assumption,

$$(4) \quad (\text{H3, uncorrelated}) \quad \text{Cov}(e_i, e_j) = 0, \quad i \neq j.$$

Table 1 lists, for three representative papers, the property that each one takes over from the 1996 result and how it is taken over. In every case a property of the BFP quantization error is inherited from that result without being re-derived under modern formats or neural-network activation distributions.

The same inheritance appears for modern MX formats. Wu et al. [26] note that “existing work treats the quantization error as a monolithic noise term”, while their own three-way decomposition of the MXFP4 error (scale bias, dead zone, grid noise) is defined in amplitude space and does not touch the conditional mean of the signed error. One citation deserves a qualification: Fan et al. [12] do cite the 1996 reference, but only in the historical sense that BFP has long been used in FPGA designs, and they attribute the CNN error analysis instead to Song, Liu, and Wang; we therefore exclude them from Table 1.

The original application context of the 1996 work matters here: audio, radar, and array signal processing, whose inputs are approximately symmetric within a block. Symmetry is therefore the substantive condition for (H1) and (H3), whereas modern neural-network activations—especially post-ReLU activations—are nonnegative, so conditional symmetry is absent.

**2.3. Progress on the numerical analysis side.** The foundations of rigorous probabilistic error analysis were laid by Higham and Mary and co-workers [19]; subsequent developments

include probabilistic summation bounds [16], deterministic and probabilistic bounds for mixed-precision algorithms [3, 4], probabilistic analyses of stochastic rounding [10, 9, 11], low-precision PDE solution [8], multiword tensor-core arithmetic and integer-splitting matrix multiplication [13, 1], mixed-precision iterative refinement [6], BFP acceleration of CNNs [12], and tooling for low-precision emulation [5].

All of these works rely on (H1) and on independence, or at least on uncorrelatedness; but none of the objects analysed contains a data-determined shared exponent. In (1) the exponent  $\xi$  is a function of all elements of the block, which makes the within-block errors intrinsically coupled through  $\xi$  (contrast Sao et al. [23] and Fox–Lindstrom [14]).

### 3. Formalization and rigorous results.

**3.1. Notation.** Let the elements of a block be i.i.d. with finite error second moments, and let

$$(5) \quad m(\xi) = \mathbb{E}[e_1 \mid \xi], \quad c(\xi) = \text{Cov}(e_i, e_j \mid \xi) \ (i \neq j), \quad \rho = \frac{\text{Cov}(e_i, e_j)}{\text{Var}(e)} \ (i \neq j).$$

**LEMMA 1** (Conditional exchangeability holds automatically). *If the block elements  $x_1, \dots, x_n$  are i.i.d. and  $\xi$  is a symmetric function of  $(|x_1|, \dots, |x_n|)$ , then  $(e_1, \dots, e_n)$  is exchangeable given  $\xi$ ; consequently  $m(\xi)$  and  $c(\xi)$  do not depend on the indices, and  $\rho$  does not depend on which pair  $i \neq j$  is chosen.*

*Proof.* For any permutation  $\pi$ ,  $(x_{\pi(1)}, \dots, x_{\pi(n)})$  has the same distribution as  $(x_1, \dots, x_n)$ , and  $\xi$  in (1) is invariant under permutations, so both have the same conditional law given  $\xi$ . Since  $e_i$  depends on the data only through  $(x_i, \xi)$ , the vectors  $(e_{\pi(1)}, \dots, e_{\pi(n)})$  and  $(e_1, \dots, e_n)$  are identically distributed given  $\xi$ .  $\square$

The variance of the within-block error sum is

$$(6) \quad \text{Var} \left[ \sum_{i=1}^n e_i \right] = n \text{Var}(e) \underbrace{(1 + (n-1)\rho)}_{=: \mathcal{I}_n}.$$

We call  $\mathcal{I}_n$  the *variance inflation factor*, the central quantity of this paper: the classical model (H1)–(H3) implies  $\mathcal{I}_n = 1$  (and conversely, only  $\text{Cov}(e_i, e_j) = 0$  is needed, not zero means). Conditional exchangeability, often listed as an additional hypothesis, follows from i.i.d. inputs and symmetry of  $\xi$  by Lemma 1, so the only hypothesis left to test is *conditional uncorrelatedness* (A2), measured in subsection 5.6.

### 3.2. Covariance decomposition.

**PROPOSITION 2** (Covariance decomposition). *Let the within-block errors  $(e_1, \dots, e_n)$  be exchangeable given  $\xi$  and have finite second moments. Then, without any additional assumption,*

$$(7) \quad \text{Cov}(e_i, e_j) = \mathbb{E}[c(\xi)] + \text{Var}(m(\xi)), \quad i \neq j.$$

*If in addition the conditional uncorrelatedness assumption*

$$(8) \quad (\text{A2}) \quad \text{Cov}(e_i, e_j \mid \xi) = 0, \quad i \neq j, \quad \text{i.e.} \quad c(\xi) \equiv 0,$$

*holds, then (7) reduces to  $\text{Cov}(e_i, e_j) = \text{Var}(m(\xi))$ , and*

$$(9) \quad \rho = \frac{\text{Var}(m(\xi))}{\text{Var}(e)}, \quad \mathcal{I}_n = 1 + (n-1) \frac{\text{Var}(m(\xi))}{\text{Var}(e)}.$$

*Proof.* By the law of total covariance,  $\text{Cov}(e_i, e_j) = \mathbb{E}[\text{Cov}(e_i, e_j \mid \xi)] + \text{Cov}(\mathbb{E}[e_i \mid \xi], \mathbb{E}[e_j \mid \xi])$ . The first term is  $\mathbb{E}[c(\xi)]$ ; since  $\mathbb{E}[e_i \mid \xi] = \mathbb{E}[e_j \mid \xi] = m(\xi)$  by Lemma 1, the second is  $\text{Cov}(m(\xi), m(\xi)) = \text{Var}(m(\xi))$ . Equation (9) follows from the definition of  $\rho$  and (6).  $\square$

COROLLARY 3 (Rigorous bounds on the inflation factor). *Always,*

$$(10) \quad 0 \leq \mathcal{I}_n \leq n, \quad \text{that is,} \quad \text{Var} \left[ \sum_{i=1}^n e_i \right] \leq n^2 \text{Var}(e).$$

*Proof.* The left inequality is  $\text{Var}[\sum_i e_i] = n \text{Var}(e) + n(n-1) \text{Cov}(e_i, e_j) \geq 0$ , equivalent to  $\rho \geq -1/(n-1)$ . The right one follows from Cauchy–Schwarz:  $e_i$  and  $e_j$  are identically distributed, so  $\text{Cov}(e_i, e_j) \leq \sqrt{\text{Var}(e_i) \text{Var}(e_j)} = \text{Var}(e)$ , hence  $\rho \leq 1$ .  $\square$

*Remark 4.* The bound  $\mathcal{I}_n \leq n$  is substantive: the within-block error sum has standard deviation growing at most linearly in  $n$ , so the *asymptotic* slope of  $\log \mathcal{I}_n$  against  $\log n$  is at most 1 and the local slope 1.134 measured in subsection 5.3 is necessarily a transient (subsection 3.6 sharpens this to  $\mathcal{I}_n/n \rightarrow 0$ ). It also flags impossible outputs:  $\mathcal{I}_n < 0$  or  $\mathcal{I}_n > n$  must be wrong. Conversely, Proposition 2 attributes the within-block correlation entirely to the fluctuation of  $m(\xi)$  and the mean of  $c(\xi)$ : if  $m(\xi) \equiv 0$  and  $c(\xi) \equiv 0$ —which subsection 5.5 proves for stochastic rounding—then (H3) holds exactly and  $\mathcal{I}_n \equiv 1$  however large  $\text{Var}(e)$  may be, whereas a nondegenerate  $m(\xi)$  gives *every* pair the same nonzero covariance, accumulating like  $n^2$  rather than  $n$  in a summation.

Using  $e_i = s \cdot (\text{RN}(x_i/s) - x_i/s)$ , the conditional mean is

$$(11) \quad m(\xi) = s \cdot \delta(\xi), \quad \delta(\xi) = \mathbb{E}[\text{RN}(u) - u \mid \xi], \quad u = x/s,$$

where  $\delta(\xi)$  is the mean unit-step rounding residual given  $\xi$ ;  $\delta(\xi) = 0$  whenever the conditional distribution of  $u$  is symmetric about 0.

### 3.3. An exact counterpart for stochastic rounding.

PROPOSITION 5 (Zero inflation under stochastic rounding). *Let the quantization use stochastic rounding (SR): given  $x$  and  $s$ , put  $u = x/s$ ,  $k = \lfloor u \rfloor$  and  $\theta = u - k \in [0, 1)$ , and take the values  $k$  and  $k+1$  with probabilities  $1-\theta$  and  $\theta$ , respectively, with the randomizations mutually independent given  $(x_1, \dots, x_n)$ . Then  $\mathbb{E}[e_i \mid x, \xi] = 0$ ,  $m(\xi) \equiv 0$ ,  $\text{Cov}(e_i, e_j) = 0$  ( $i \neq j$ ) and  $\mathcal{I}_n \equiv 1$ .*

*Proof.* Given  $x$  and  $s$ , the definition of SR gives  $\mathbb{E}[\text{RN}(u)] = k(1-\theta) + (k+1)\theta = k + \theta = u$ , hence  $\mathbb{E}[e \mid x, s] = s(\mathbb{E}[\text{RN}(u)] - u) = 0$  and, by conditional expectation,  $m(\xi) = \mathbb{E}[\mathbb{E}[e \mid x, \xi] \mid \xi] = 0$ . Since the SR randomizations of the individual elements are mutually independent given  $(x_1, \dots, x_n)$ ,  $\text{Cov}(e_i, e_j \mid \xi, x) = 0$ ; the law of total covariance then gives  $\text{Cov}(e_i, e_j \mid \xi) = 0$ , because both  $\mathbb{E}[e_i \mid x, \xi]$  and  $\mathbb{E}[e_j \mid x, \xi]$  vanish. Hence (A2) holds automatically for SR, and Proposition 2 gives  $\text{Cov}(e_i, e_j) = \text{Var}(m(\xi)) = 0$ .  $\square$

COROLLARY 6 (Variance law under stochastic rounding). *Write  $\theta = \text{frac}(x/s) \in [0, 1)$ . Under stochastic rounding,*

$$(12) \quad \mathbb{E}[e^2 \mid \xi] = s^2 \mathbb{E}[\theta(1-\theta) \mid \xi], \quad \text{Var}(e) = \mathbb{E}[s^2 \theta(1-\theta)].$$

*In particular, if  $\theta$  is conditionally uniform on  $[0, 1)$  given  $\xi$ , then  $\text{Var}(e) = \mathbb{E}[s^2]/6$ , exactly twice the classical round-to-nearest value  $\mathbb{E}[s^2]/12$ .*

*Proof.* Given  $x$  (hence  $u$  and  $\theta$ ),  $\text{RN}(u)$  takes the values  $k, k+1$  with probabilities  $1-\theta, \theta$ , so  $g = \text{RN}(u) - u$  equals  $-\theta$  with probability  $1-\theta$  and  $1-\theta$  with probability  $\theta$ , giving  $\mathbb{E}[g^2 \mid x, s] = \theta(1-\theta)$  and  $\mathbb{E}[g \mid x, s] = 0$ . Multiplying by  $s^2$  and taking conditional expectations gives the first identity, and the second follows from  $\text{Var}(e) = \mathbb{E}[e^2]$  since  $m \equiv 0$ . If  $\theta \mid \xi$  is uniform,  $\mathbb{E}[\theta(1-\theta)] = 1/6$ , so  $\text{Var}(e) = \mathbb{E}[s^2]/6$ .  $\square$

Proposition 5 provides an operational probe: in a shared-exponent block, stochastic rounding drives the unobservable  $m(\xi)$  to zero, so if the inflation in subsection 5.3 is caused by  $m(\xi) \neq 0$ ,

switching to SR must annihilate it exactly, independently of the distribution, block length, and mantissa width. Subsection 5.5 carries out this test.

**3.4. Exact variance laws and the reduced model.** Both  $m(\xi)$  and the variance structure reduce to one-dimensional functions, turning the scaling question into a computable object. With  $g(u) = \text{RN}(u) - u \in [-\frac{1}{2}, \frac{1}{2}]$ ,  $e_i = s g(x_i/s)$ .

PROPOSITION 7 (Variance law). Write  $\gamma(\xi) = \mathbb{E}[g(x_1/s)^2 \mid \xi]$  and  $\delta(\xi) = \mathbb{E}[g(x_1/s) \mid \xi] = m(\xi)/s$ . Then

$$(13) \quad \mathbb{E}[e^2] = \mathbb{E}[s^2 \gamma(\xi)], \quad \text{Var}(e) = \mathbb{E}[s^2 \gamma(\xi)] - (\mathbb{E}[e])^2.$$

The classical model (H1)–(H2) is exactly equivalent to  $\delta \equiv 0$  and  $\gamma \equiv 1/12$ .

*Proof.* Taking expectations in  $e^2 = s^2 g(x/s)^2$  (with  $s$  a function of  $\xi$ ) gives the first identity; the second follows from  $\text{Var}(e) = \mathbb{E}[e^2] - (\mathbb{E}[e])^2$ . Moreover  $\mathbb{E}[e^2] = \mathbb{E}[s^2/12]$  if and only if  $\gamma \equiv 1/12$ , and  $\mathbb{E}[e] = 0$  if and only if  $\delta \equiv 0$ .  $\square$

Proposition 7 shows that the failure of the classical variance and the emergence of correlation are *two independent structural defects*, the former  $\gamma \not\equiv 1/12$  and the latter  $\delta \not\equiv 0$ , so the deviations of subsection 5.2 and subsection 5.3 cannot mask each other.

The following lemma shows that, given  $\xi$ , the conditional distribution enters the behavior of  $g$  only through the step  $s$ , with error controlled by the distance of the block maximum from the bulk; this legitimizes the reduced model below.

LEMMA 8 (Two-piece mixture representation of the conditional law). Write  $a = 2^\xi$ ,  $b = 2^{\xi+1}$ ,  $F_a = F(a)$ ,  $F_b = F(b)$  and  $P_k = \mathbb{P}(\xi = k) = F_b^n - F_a^n$ . Given  $\xi = k$ , the conditional law of  $x_1$  is a mixture of two truncated distributions,

$$(14) \quad \mathbb{P}(x_1 \in A \mid \xi = k) = w_A \mathbb{P}_F(x_1 \in A \mid |x_1| < a) + w_B \mathbb{P}_F(x_1 \in A \mid a \leq |x_1| < b),$$

with  $w_A = F_a(F_b^{n-1} - F_a^{n-1})/P_k$ ,  $w_B = F_b^{n-1}(F_b - F_a)/P_k$  and  $w_A + w_B = 1$ . Consequently, for every bounded measurable  $\varphi$ ,

$$(15) \quad |\mathbb{E}[\varphi(x_1) \mid \xi = k] - \mathbb{E}_F[\varphi]| \leq 2\|\varphi\|_\infty [w_A(1 - F_a) + w_B(1 - F_b + F_a)].$$

If the support of  $F$  is unbounded, then as  $k \rightarrow \infty$  we have  $F_a, F_b \rightarrow 1$ ,  $w_A \rightarrow (n-1)/n$  and  $w_B \rightarrow 1/n$ , so the right-hand side of (15) tends to 0.

*Proof.* Given  $x_1 = x$  with  $|x| < a$ , the event  $\{\xi = k\}$  is equivalent to  $\max_{j \geq 2} |x_j| \in [a, b)$ , of probability  $F_b^{n-1} - F_a^{n-1}$ ; given  $a \leq |x| < b$  it is equivalent to  $\max_{j \geq 2} |x_j| < b$ , of probability  $F_b^{n-1}$ . Multiplying these by  $f(x)/P_k$  yields (14), whose normalization follows from  $F_a(F_b^{n-1} - F_a^{n-1}) + F_b^{n-1}(F_b - F_a) = F_b^n - F_a^n$ . The error bound follows from  $\mathbb{E}[\varphi \mid \xi = k] - \mathbb{E}_F[\varphi] = \sum_{i \in \{A, B\}} w_i (\mathbb{E}_F[\varphi \mid A_i] - \mathbb{E}_F[\varphi])$  together with  $|\mathbb{E}_F[\varphi \mid A_i] - \mathbb{E}_F[\varphi]| \leq 2\|\varphi\|_\infty (1 - \mathbb{P}_F(A_i))$ , where  $1 - \mathbb{P}_F(A) = 1 - F_a$  and  $1 - \mathbb{P}_F(B) = 1 - F_b + F_a$ .  $\square$

*Remark 9.* Equation (15) makes precise when “the conditional law enters only through the step” is valid: the deviation is controlled by two *truncations*, at  $a$  and at  $b$ . It degenerates when  $2^\xi$  lies far below the bulk scale of the data ( $F_a \rightarrow 0$ )—substantive information, not a technical defect: almost all elements then fall inside the rounding dead zone and the conditional law genuinely differs from the unconditional one, its response governed by the other asymptotics (19). Conversely, as  $2^\xi$  grows the truncations disappear, so the reduced model of subsection 5.7 is more accurate on the *large*- $\xi$  side and Proposition 12 rests on the same limit. Where the right-hand side of (15) is negligible,  $\varphi(x) = g(x/s)$  (with  $\|\varphi\|_\infty = \frac{1}{2}$ ) gives

$$(16) \quad \begin{aligned} \delta(\xi) &= \delta(s) + O(\varepsilon), & \gamma(\xi) &= \gamma(s) + O(\varepsilon), \\ \delta(s) &= \mathbb{E}_F[\text{RN}(x/s) - x/s], & \gamma(s) &= \mathbb{E}_F[(\text{RN}(x/s) - x/s)^2], \end{aligned}$$

with  $\varepsilon$  that right-hand side; over the experimental regime this remainder has size  $10^{-3}$ , which is why the reduced model reproduces the simulations to three or four significant digits.

**Reduced model.** Retaining the leading term of (16) yields the central claim: every statistic of the within-block error is determined by two one-dimensional functions of the distribution and the step,

$$(17) \quad m(\xi) = s \delta(s), \quad \mathbb{E}[e^2 \mid \xi] = s^2 \gamma(s), \quad s = 2^{\xi - L_m},$$

so that

$$(18) \quad \begin{aligned} \text{Var}(e) &= \mathbb{E}[s^2 \gamma(s)] - (\mathbb{E}[s \delta(s)])^2, & \text{Cov}(e_i, e_j) &= \text{Var}(s \delta(s)), \\ \rho &= \frac{\text{Var}(s \delta(s))}{\text{Var}(e)}, & \mathcal{I}_n &= 1 + (n-1) \frac{\text{Var}(s \delta(s))}{\text{Var}(e)}. \end{aligned}$$

In (17) and (18) the only random input is the shared step  $s$  (the rounded logarithmic scale of  $M_n$ ), while  $\delta$  and  $\gamma$  follow from one-dimensional quadrature: the problem factorizes into a *deterministic response curve* and an *order statistic* (the distribution of  $M_n$ ). Three consequences follow.

- The symmetry criterion is exact: if  $F$  is symmetric about 0, then  $g(\cdot/s)$  is odd, so  $\delta(s) \equiv 0$ , hence  $\text{Cov}(e_i, e_j) = 0$  and  $\mathcal{I}_n \equiv 1$  exactly, independently of (A2).
- The scaling question reduces to an extreme-value question: the growth rate of  $\mathcal{I}_n - 1$  is governed entirely by whether  $B_n = \text{Var}(s \delta(s))$  decays with  $n$ , and the law of  $s$  depends on  $n$  only through  $M_n$ ; with Corollary 3 this bounds the exponent by 1, and Proposition 12 in subsection 3.6 yields  $B_n \rightarrow 0$ .
- The  $L_m$  dependence has a quantitative explanation. In the “dead zone” regime (step below the bulk scale of the data) most elements of a block satisfy  $|x_i| < s/2$  and round to 0; a direct computation gives  $\delta(s) = -\alpha s + o(s)$  and  $\gamma(s) = \beta + o(1)$ , with  $\alpha, \beta > 0$  depending only on the shape of  $F$  near 0. Then  $m = s\delta \propto s^2$  and  $B_n \propto \text{Var}(s^2)$ , whereas  $\text{Var}(e) \approx \mathbb{E}[s^2 \gamma] \propto \mathbb{E}[s^2]$ ; since one more bit of  $L_m$  halves  $s$ ,

$$(19) \quad \text{Var}(s^2) \rightarrow \frac{1}{16} \text{Var}(s^2), \quad \mathbb{E}[s^2] \rightarrow \frac{1}{4} \mathbb{E}[s^2], \quad \rho \propto 4^{-L_m}, \quad \mathcal{I}_n - 1 \propto 4^{-L_m}.$$

The data of subsection 5.7 support this scaling (16.3 for ReLU, 16.9 for the half-normal, close to  $4^2 = 16$ ); it fails near the peak of  $|\delta(s)|$ , which explains the non-monotone  $L_m$  dependence in subsection 5.3.

**3.5. The two regimes of the support.** The first regime is Proposition 10; the second is Proposition 12 in subsection 3.6, where the inflation is transient.

**PROPOSITION 10** (Bounded support: exponential decay). *Suppose  $|x_1|$  has bounded support,  $B = \text{ess sup } |x_1| < \infty$ , and  $\mathbb{E}[\text{Var}(e \mid \xi)] > 0$ . Let  $\xi_0$  be the integer with  $2^{\xi_0} < B \leq 2^{\xi_0+1}$  and write  $p = F(2^{\xi_0}) < 1$ . Then  $\xi \leq \xi_0$  almost surely, and*

$$(20) \quad \begin{aligned} \mathbb{P}(\xi \neq \xi_0) &\leq p^n, & \text{Var}(m(\xi)) &\leq C p^n, \\ \mathcal{I}_n - 1 &\leq \frac{(n-1) C p^n}{\mathbb{E}[\text{Var}(e \mid \xi)] (1 - p^n)} \rightarrow 0, \end{aligned}$$

where  $C$  depends only on  $F$  and  $L_m$ , and the convergence is exponentially fast.

*Proof.* Since  $M_n \leq B \leq 2^{\xi_0+1}$ , we have  $\xi \leq \xi_0$  except on the null event  $\{M_n = B = 2^{\xi_0+1}\}$ . Moreover  $\{\xi \neq \xi_0\} \subseteq \{\xi \leq \xi_0 - 1\} \subseteq \{M_n \leq 2^{\xi_0}\}$ , so  $\mathbb{P}(\xi \neq \xi_0) \leq F(2^{\xi_0})^n = p^n$ . From  $|g| \leq 1/2$  we get  $|m(\xi)| \leq \frac{1}{2} 2^{\xi_0 - L_m}$ , hence  $\text{Var}(m) \leq (\max m - \min m)^2 \mathbb{P}(\xi \neq \xi_0) \leq C p^n$ . Finally  $\text{Var}(e) = \mathbb{E}[\text{Var}(e \mid \xi)] + \text{Var}(m)$ , and substitution into (9) gives the claim.  $\square$

*Remark 11.* Proposition 10 is meaningful only in the nondegenerate case  $\mathbb{E}[\text{Var}(e \mid \xi)] > 0$ . The conclusion is *independent of the endpoint tail index*: bounded support forces  $\xi$  onto a single grid point, so the inflation must vanish exponentially; heavy tails accompany unbounded support and are thus not necessary for the inflation. The quantity  $p = F(2^{\xi_0})$  in (20) is often close to 1 for continuous  $F$ , making the bound conservative: the real mechanism is concentration of  $\xi$  on the *mode* grid point (subsection 5.9 reports numerically that  $\mathbb{P}(\xi = \text{mode})$  already reaches 0.97–1.00 at  $n = 4096$ ).

**3.6. Unbounded support: the transient nature of the inflation.** With unbounded support the inflation also disappears, but so slowly that over attainable block lengths it looks like a power law.

PROPOSITION 12 (Asymptotic disappearance of the inflation under unbounded support). *Suppose  $|x_1|$  has unbounded support,  $\mathbb{E}[x_1^2] < \infty$  and  $\text{Var}(x_1) > 0$ . Write  $A_n = \mathbb{E}[\text{Var}(e \mid \xi)]$  and  $B_n = \text{Var}(m(\xi))$ , so that  $\text{Var}(e) = A_n + B_n$ . Then, as  $n \rightarrow \infty$ ,*

$$(21) \quad m(\xi) \longrightarrow -\mathbb{E}[x_1] \quad \text{a.s.}, \quad B_n \longrightarrow 0, \quad A_n \longrightarrow \text{Var}(x_1) > 0,$$

and hence, under assumption (A2),

$$(22) \quad \rho_n = \frac{B_n}{A_n + B_n} \longrightarrow 0, \quad \mathcal{I}_n = 1 + (n-1)\rho_n = \frac{A_n + nB_n}{A_n + B_n}, \quad \frac{\mathcal{I}_n}{n} \leq \frac{1}{n} + \frac{B_n}{A_n} \longrightarrow 0.$$

In particular, no unbounded distribution with finite second moment yields  $\mathcal{I}_n \asymp n$ : linear accumulation is at best a transient over moderate block lengths.

*Proof.* Write  $s = 2^{\xi - L_m}$  and  $D = \{|x_1| < s/2\}$  (the dead zone). By Lemma 8, given  $\xi = k$  the conditional law of  $x_1$  is a mixture of the truncated components  $A = \{|x_1| < 2^k\}$  and  $B = \{2^k \leq |x_1| < 2^{k+1}\}$  with weights  $w_A \rightarrow (n-1)/n$  and  $w_B \rightarrow 1/n$  as  $k \rightarrow \infty$ .

*Component B.* On it  $|x_1| \geq 2^k = s2^{L_m}$ , so  $w_B \leq \mathbb{E}[|x_1|; 2^k \leq |x_1| < 2^{k+1}]/2^k \rightarrow 0$  (because  $\mathbb{E}|x_1| < \infty$ ); and on it  $|e_1| \leq s/2$ , so the component contributes at most  $sw_B/2 \leq \mathbb{E}[|x_1|; |x_1| \geq 2^k]/2^{L_m+1} \rightarrow 0$ .

*Component A.* There  $\mathbb{P}(D^c \mid A) \leq \bar{F}(s/2)/F(2^k) \rightarrow 0$ , and on  $D^c$  we have  $|e_1| \leq s/2$ , so the contribution is at most  $(s/2)\bar{F}(s/2)/F(2^k) \rightarrow 0$  (because  $t\bar{F}(t) \rightarrow 0$ ); on  $D$ ,  $e_1 = -x_1$  with truncation bound  $s/2 \rightarrow \infty$ , so  $\mathbb{E}[e_1 \mid D \mid A] \rightarrow -\mathbb{E}[x_1]$ .

Combining the components gives  $m(\xi = k) \rightarrow -\mathbb{E}[x_1]$  (the exact limit is  $-(1 - n^{-1})\mathbb{E}[x_1]$ ). By the estimates above,  $|m(\xi)| \leq w_A\mathbb{E}[|x_1| \mid A] + \frac{s}{2}w_B \leq 2\mathbb{E}|x_1|$  is uniformly bounded, so dominated convergence gives  $\mathbb{E}[m(\xi)] \rightarrow -\mathbb{E}[x_1]$  and  $\mathbb{E}[m(\xi)^2] \rightarrow (\mathbb{E}[x_1])^2$ , that is,  $B_n \rightarrow 0$ . Replacing  $\varphi$  by  $g(x/s)^2$  and repeating the argument gives  $A_n \rightarrow \text{Var}(x_1)$ . Finally, the law of total variance with  $\rho_n = B_n/(A_n + B_n)$  from Proposition 2, substituted into (6), yields (22).  $\square$

*Remark 13* (Peak and fall-off: numerical evidence from the model). Proposition 12 gives only the limit, not the rate. Evaluating the right-hand side of (18) *exactly* by one-dimensional quadrature (not Monte Carlo; Pareto(4),  $L_m = 2$ , 60 decimal digits to eliminate cancellation error) yields the whole population curve of  $\mathcal{I}_n$ : it peaks at  $5.9 \times 10^4$  near  $n \approx 10^5$ , then falls monotonically back to 1 (52 at  $n = 10^{12}$ , 1.002 at  $n = 10^{20}$ ), and reproduces the measured values (Table 9):  $3.61 \times 10^4$  at  $n = 10^6$  and  $9.2 \times 10^3$  at  $n = 10^7$ , against measured  $3.61 \times 10^4$  and  $9.6 \times 10^3$ . The fall-off is therefore *population* behavior, not a sampling artifact (see subsection 5.8). The dependence on  $n$  is also not smooth: rounding in  $\xi = \lfloor \log_2 M_n \rfloor$  makes  $B_n$  oscillate with  $n$ , so a power-law fit over a finite range is inappropriate.

*Remark 14* (What remains open). Equation (22) gives  $\mathcal{I}_n = o(n)$  but not  $\mathcal{I}_n \rightarrow 1$ : the latter is equivalent to  $nB_n \rightarrow 0$  and would require an *n-independent* regularity assumption on  $\delta(s)$  as  $s \rightarrow \infty$ , which we do not state here. Moreover, for  $\alpha \leq 4$  the fourth moment of  $e$  diverges, so the sample estimator of  $\mathcal{I}_n$  has no finite variance; the limit (21) is therefore accessible only

TABLE 2  
The six distributions used in the experiments.

| Distribution   | Symmetry   | Support and rationale                                                            |
|----------------|------------|----------------------------------------------------------------------------------|
| Gaussian       | symmetric  | unbounded; baseline and control                                                  |
| Student- $t_3$ | symmetric  | unbounded; heavy-tailed control (separates heavy tails from symmetry)            |
| Spike mixture  | symmetric  | unbounded; outlier channel $95\% \mathcal{N}(0, 0.01^2) + 5\% \mathcal{N}(0, 1)$ |
| Lognormal      | asymmetric | unbounded; strongly asymmetric, heavy right tail                                 |
| Half-normal    | asymmetric | bounded on one side; intermediate case                                           |
| ReLU output    | asymmetric | bounded on one side; the standard case for neural-network activations            |

by high-precision evaluation of the model, not “verified” by simulation with a fixed number of blocks.

#### 4. Numerical experiment design.

**4.1. Distributions and format parameters.** The experiments use the six distributions listed in Table 2, chosen to separate the two factors “symmetry” and “heavy tails”:

The mantissa widths are  $L_m = 2, 3, 4, 7$ , corresponding to E5M2, E4M3/MXFP4, 4-bit and bf16 scale, respectively. The shared exponent and the step are computed as in (1), and rounding is round-to-nearest (ties to even) unless stated otherwise. All experiments use fixed random seeds. The reproduction scripts, the raw outputs of the runs reported here, and a table mapping each script to the results it produces are openly available at <https://github.com/NUAApanda-1116/bfp-error-independence>.

**4.2. The estimator and the sequence of corrections.** By (6),  $\mathcal{I}_n$  can be recovered from  $\rho$  or estimated directly. Our estimator went through three rounds of correction, recorded here because the first two are instructive errors.

*Two rejected rounds.* The initial implementation computed the within-block correlation after centring within each block, which forces the within-block error sum to zero and hence makes the average pairwise correlation *necessarily* equal to  $-1/(n-1)$ : across 48 configurations the estimate depended only on the block size ( $-1/15$ ,  $-1/31$ ,  $-1/63$ ) and not at all on the distribution. The second version estimated the average covariance from a single pair  $(e_0, e_1)$  and formed  $1 + (n-1)\rho$ ; its standard error is inflated by the factor  $(n-1)$ , and it produced *negative* inflation factors at  $n = 256$  and  $n = 512$  ( $-17.487$  and  $-6.797$ ), whereas  $\mathcal{I}_n \geq 0$  always holds. We therefore withdraw all scaling conclusions that rested on it.

*The estimator used here.* Rather than estimating  $\rho$ , we estimate the quantity in (6) directly,

$$(23) \quad \widehat{\mathcal{I}}_n = \frac{\widehat{\text{Var}}_{\text{blocks}}(\sum_{i \in \text{block}} e_i)}{n \cdot \widehat{\text{Var}}_{\text{elem}}(e)},$$

whose numerator and denominator are both low-variance statistics, using all block sums and all elements respectively and depending on no single pair. Error bars are standard errors over 20 independent groupings of 1000 blocks each; all data in section 5 come from this estimator and satisfy  $\mathcal{I}_n \geq 0$ .

*A fourth tool.* To measure the two terms of Proposition 2 separately (subsections 5.6 and 5.7) a further script (`bfp_error_pilot.v7.py`) bins by the integer value of  $\xi$ , centres by the *within-bin (conditional) mean*  $m_b$ , and retains only  $(S, \sum_i e_i^2, \xi, s)$  per block, which by the

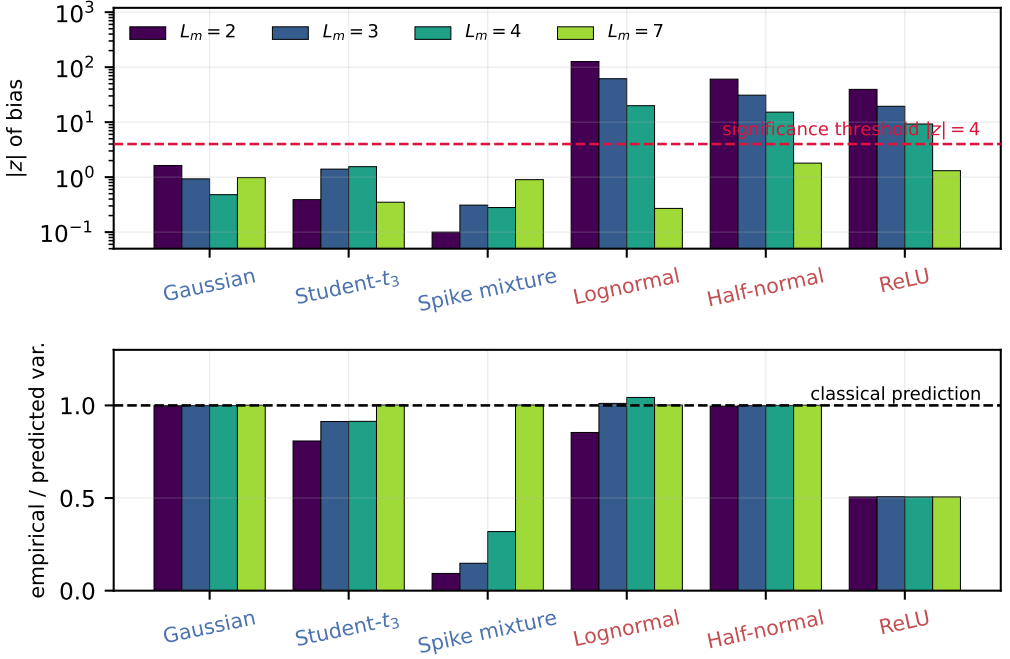


FIG. 1. *Top: significance  $|z|$  of the block-mean bias (logarithmic vertical axis); the red dashed line is the threshold  $|z| = 4$ , blue axis labels denote symmetric distributions and red ones asymmetric distributions. Bottom: ratio of the measured variance to the classical prediction  $\mathbb{E}[s^2]/12$ ; the dashed line is the classical prediction 1. In both panels all four mantissa widths lie below the threshold for the symmetric distributions, and all asymmetric distributions are significant for  $L_m \leq 4$ . The ReLU variance ratio is constantly about 0.506; for the spike mixture it drops to 0.093 at low precision (the classical formula overestimates by about a factor of 10).*

identity  $\sum_{i \neq j} (e_i - m_b)(e_j - m_b) = (S - nm_b)^2 - \sum_i (e_i - m_b)^2$  (with  $S = \sum_i e_i$ ) determines  $c(\xi)$  exactly. The centring is essential: centring by the within-block mean makes the average pairwise covariance necessarily equal to  $-\text{Var}(e)/(n-1)$ , the artefact of the first round, whereas centring by the within-bin mean estimates  $\text{Cov}(e_i, e_j | \xi)$  directly. The same script yields  $\mathcal{I}_n$  along three independent paths (subsection 5.6), an independent check of the data in section 5.

**5. Results. Data provenance.** The tables of this section come from several batches of runs, each drawing its own random blocks, and the archived terminal output of every batch is distributed with the reproduction material together with the mapping from table to output file. Comparisons should therefore be made *within* a table rather than across tables at a precision finer than a few parts in a thousand: the same nominal configuration can differ by more than one standard error, as for ReLU at  $L_m = 2$ ,  $n = 512$ , where the value is 1.124 in Table 4 and 1.110 in Table 6, about 0.9%. None of the qualitative conclusions of this section depends on a difference of that size.

**5.1. Failure of the zero-mean assumption.** We test (H1) with a  $z$  statistic on the block mean:  $N = 4 \times 10^4$  blocks and  $z = \bar{e}_{\text{block}}/(\hat{\sigma}_{\text{block}}/\sqrt{N})$ . Figure 1 (top) gives the results. All symmetric distributions pass (H1) with  $|z| \leq 1.63$ , whereas all asymmetric ones fail for  $L_m \leq 4$  ( $z = -126.80, -60.57$  and  $-39.48$  at  $L_m = 2$  for the lognormal, half-normal and ReLU respectively, and  $-19.92, -15.22, -9.26$  at  $L_m = 4$ ), and the bias grows as the mantissa width decreases. The relative bias (bias divided by the mean magnitude) reaches  $-6.19\%$  for the lognormal at  $L_m = 2$ . Values of  $|z|$  for  $L_m = 3$  are  $-61.69, -30.96$  and  $-19.45$ ; all figures are those plotted in Figure 1 (top), computed from  $N = 4 \times 10^4$  blocks.

TABLE 3  
Ratio of the measured variance to the classical prediction  $\mathbb{E}[s^2]/12$ .

| Distribution   | $L_m = 2$    | $L_m = 3$    | $L_m = 4$    | $L_m = 7$    |
|----------------|--------------|--------------|--------------|--------------|
| Gaussian       | 0.999        | 0.999        | 0.999        | 1.001        |
| Student- $t_3$ | 0.808        | 0.913        | 0.914        | 1.002        |
| Spike mixture  | <b>0.093</b> | <b>0.148</b> | <b>0.319</b> | 1.002        |
| Lognormal      | 0.854        | 1.011        | 1.043        | 1.002        |
| Half-normal    | 0.996        | 0.999        | 1.000        | 1.001        |
| ReLU           | <b>0.506</b> | <b>0.507</b> | <b>0.506</b> | <b>0.506</b> |

The mechanism agrees with (11): the expected round-to-nearest error vanishes for symmetric inputs, whereas for nonnegative inputs the opportunities to round down and up are unequal and a systematic negative bias appears. Post-ReLU activations are exactly nonnegative—not a special case, but the norm in the forward pass of modern neural networks.

**5.2. Structural bias of the classical variance formula.** Taking  $\mathbb{E}[s^2]/12$  as the prediction and the measured variance as the comparison, the ratios are given in Table 3 and Figure 1 (bottom).

By Proposition 7, the ratio of the measured to the classical variance has the closed form

$$(24) \quad \frac{\text{Var}(e)}{\mathbb{E}[s^2]/12} = \frac{12(\mathbb{E}[s^2\gamma(\xi)] - (\mathbb{E}[e])^2)}{\mathbb{E}[s^2]},$$

so a deviation from 1 has a single source: the departure of  $\gamma(\xi) = \mathbb{E}[g(x/s)^2 \mid \xi]$  from  $1/12$  (plus the small term  $-12(\mathbb{E}[e])^2/\mathbb{E}[s^2]$ ). The condition  $\gamma \equiv 1/12$  holds when, given  $\xi$ , the normalized residual is *conditionally uniform* within its rounding cell—assumed implicitly by the classical model, and false for neural-network activations. Three departures appear.

First, the ReLU ratio is pinned near 0.500 and does not vary with the mantissa width or the block length, because the model omits a *probability atom*: exactly half of the ReLU outputs are precisely zero and their rounding error is identically zero, so

$$(25) \quad \gamma(s) = (1 - p_0) \cdot \frac{1}{12} + p_0 \cdot 0 = \frac{1}{24}, \quad p_0 = \mathbb{P}(x = 0) = \frac{1}{2},$$

which predicts a ratio  $= 1 - p_0 = 0.500$  (that the non-atomic part still has an approximately conditionally uniform residual is consistent with the half-normal measurements). The measured value is 0.506 (Table 3,  $4 \times 10^4$  blocks), and all values over  $L_m \in \{2, 3, 4, 7\}$  and  $n \in \{32, 512, 4096\}$  lie in 0.499–0.509;  $p_0 = 1/2$  is fixed by the definition of ReLU and is not a fitted parameter.

Second, the same explanation predicts the ratio under stochastic rounding. By Corollary 6, stochastic rounding changes the within-cell variance from  $1/12$  to  $1/6$ , so

$$(26) \quad \left. \frac{\text{Var}(e)}{\mathbb{E}[s^2]/12} \right|_{\text{SR}} = 2(1 - p_0) = 1.000 \text{ (ReLU)}, \quad = 2.000 \text{ (half-normal, } p_0 = 0).$$

The data are exactly of this form: 1.015 for ReLU and 2.002 for the half-normal (from the rounding-rule runs of subsection 5.4), and 2.000 for the Gaussian in the measurement of subsection 5.6 ( $L_m = 4$ ,  $n = 4096$ ). This is a cross-validation with no parameter tuned after the fact.

Third, the value 0.093 for the spike mixture and 0.506 for ReLU come from the same mechanism:  $\gamma$  is depressed by the *dead zone*. When the step is much larger than the bulk scale of the data, most elements satisfy  $|u| < 1/2$  and round to zero, and the residual  $g(u) = -u$  is tiny; the binned data of subsection 5.6 show that 96% of the spike-mixture elements fall in the dead zone at  $n = 4096$ , where the within-bin average of  $\gamma$  is only 0.0045. In unified form: the

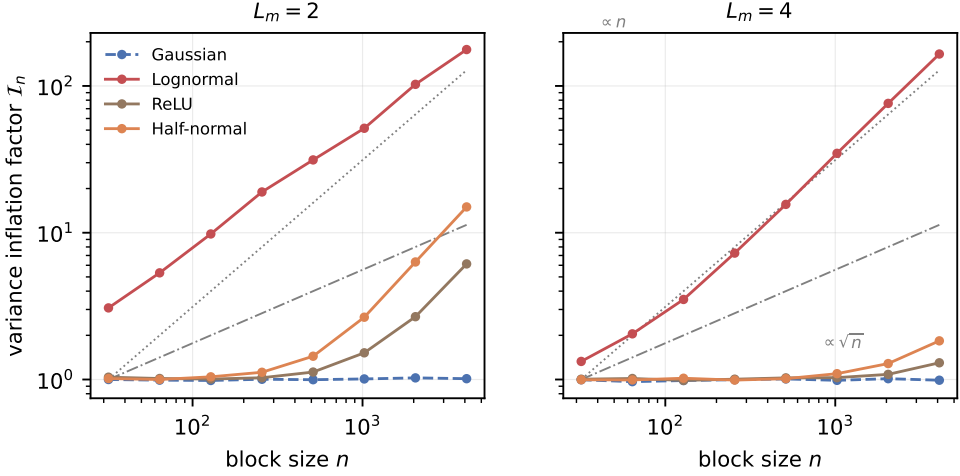


FIG. 2. Variance inflation factor  $\mathcal{I}_n$  as a function of the block length  $n$  (log-log axes). Left:  $L_m = 2$ ; right:  $L_m = 4$ . The grey dotted line is  $\propto n$  and the grey dash-dotted line is  $\propto \sqrt{n}$ . The symmetric control (Gaussian, dashed) is identically 1 for all  $n \leq 4096$ . The asymmetric distributions grow systematically with  $n$ , and the lognormal reaches a tail scaling exponent of 1.134 at  $L_m = 4$ .

TABLE 4

Variance inflation factor  $\mathcal{I}_n$  (20 groups  $\times$  1000 blocks). The last column is the local log-slope over the largest measured interval ( $n \in [512, 4096]$  for  $L_m = 2$ ,  $n \in [2048, 4096]$  for  $L_m = 4$ ).

| $L_m$ | Distribution       | 32    | 128   | 512    | 1024   | 2048    | 4096           | Tail slope  |
|-------|--------------------|-------|-------|--------|--------|---------|----------------|-------------|
| 2     | Gaussian (control) | 1.000 | 0.985 | 0.997  | 1.009  | 1.026   | 1.013          | $\approx 0$ |
| 2     | Lognormal          | 3.075 | 9.809 | 31.341 | 51.480 | 102.555 | <b>176.996</b> | 0.833       |
| 2     | ReLU               | 1.037 | 1.007 | 1.124  | 1.520  | 2.679   | <b>6.132</b>   | 0.816       |
| 2     | Half-normal        | 1.015 | 1.043 | 1.441  | 2.657  | 6.321   | <b>14.989</b>  | 1.126       |
| 4     | Gaussian (control) | 0.996 | 0.984 | 1.009  | 0.987  | 1.013   | 0.989          | $\approx 0$ |
| 4     | Lognormal          | 1.328 | 3.511 | 15.601 | 34.748 | 76.063  | <b>165.025</b> | 1.134       |
| 4     | ReLU               | 1.005 | 0.986 | 1.026  | 1.030  | 1.084   | 1.300          | 0.114       |
| 4     | Half-normal        | 0.996 | 1.020 | 1.013  | 1.096  | 1.284   | 1.834          | 0.286       |

classical formula systematically overestimates the error variance when the dead zone dominates, whereas when  $s$  is comparable to the bulk scale of the data,  $\gamma$  can slightly exceed  $1/12$  (for instance 1.043 for the lognormal at  $L_m = 4$ ) and the ratio slightly exceeds 1.

The deviations of this subsection and those of subsection 5.3 are mutually independent: by Proposition 7 the former is controlled by  $\gamma \neq 1/12$  and the latter by  $\delta \neq 0$ , and subsection 5.5 shows that stochastic rounding turns the ratio studied here into 2.000 (Corollary 6) while the inflation vanishes simultaneously, so the two must be corrected separately. One methodological caveat: both numerator and denominator of (24) are dominated by rare large- $\xi$  blocks, so convergence is slow—discarding sparse  $\xi$  bins with fewer than 20 blocks changes  $\text{Var}(e)$  by up to 4.9%, which explains the discrepancy between 0.854 in Table 3 and 0.880 in subsection 5.7 for the same nominal configuration (lognormal,  $L_m = 2$ ,  $n = 32$ ).

**5.3. Scaling behaviour of the variance inflation.** The variance inflation factor is the central quantity here. We estimate  $\mathcal{I}_n$  by (23), with the block length  $n$  increasing from 32 to 4096. Figure 2 and Table 4 give the results.

The measurements support three conclusions.

(i) (H1) and (H3) hold for symmetric distributions. For the Gaussian,  $\mathcal{I}_n = 1.000 \pm 0.01$  and stays flat in  $n$  up to  $n = 4096$ ; the same holds at every measured block length for Student- $t_3$

TABLE 5

Sensitivity to the rounding rule ( $n = 32$ ,  $L_m = 4$ ; the stochastic rounding column uses a single group of 1000 blocks, whose error bars are far larger than the  $20 \times 1000$  blocks used for the definitive conclusions, so its  $\mathcal{I}_{32}$  serves only as a qualitative reference; for the definitive test see Table 6 in subsection 5.5).

| Distribution | Rounding rule     | $z_{\text{bias}}$ | Measured/predicted variance | $\mathcal{I}_{32}$ |
|--------------|-------------------|-------------------|-----------------------------|--------------------|
| Lognormal    | nearest (RNE)     | -25.33            | 1.043                       | 1.253              |
|              | half-away         | -24.68            | 1.038                       | 1.188              |
|              | truncation        | <b>-422.71</b>    | 1.623                       | 14.270             |
|              | <b>stochastic</b> | <b>-0.98</b>      | 2.066                       | 0.825              |
| ReLU         | nearest (RNE)     | -11.29            | 0.508                       | 1.196              |
|              | half-away         | -13.16            | 0.507                       | 1.072              |
|              | truncation        | <b>-603.77</b>    | 1.315                       | 2.931              |
|              | <b>stochastic</b> | <b>-0.81</b>      | 1.015                       | 1.079              |
| Half-normal  | nearest (RNE)     | -21.08            | 0.999                       | 1.109              |
|              | half-away         | -18.23            | 1.001                       | 1.182              |
|              | truncation        | <b>-982.14</b>    | 1.155                       | 5.357              |
|              | <b>stochastic</b> | <b>0.97</b>       | 2.002                       | 1.066              |

and the spike mixture, whose  $\mathcal{I}_n$  lies in 0.99–1.01 throughout (section A). This explains why the hardware literature did not go wrong in its original, approximately symmetric setting, and it rules out implementation artefacts. It does not mean that the classical model holds as a whole: subsection 5.2 shows that the symmetric Student- $t_3$  has a variance ratio as low as 0.808 for  $L_m \leq 4$ , so (H2) can still fail. A value  $\mathcal{I}_n = 1$  only excludes correlation; it does not guarantee the variance formula.

(ii) *Growth on asymmetric distributions is unsaturated, but the exponent cannot exceed 1.* For the lognormal at  $L_m = 4$  the local log-slope over  $n \in [512, 4096]$  is 1.134, but by Corollary 3  $\mathcal{I}_n \leq n$ , so the *asymptotic* slope is at most 1: the value 1.134 can only be a transient, and extrapolating  $n^{1.13}$  would contradict the rigorous bound (subsection 3.6 proves  $\mathcal{I}_n/n \rightarrow 0$ ). The reduced model (18) gives the right reading: Table 4 yields  $\rho_n = 0.043$  at  $n = 4096$  for the lognormal, 0.0013 for ReLU and 0.0034 for the half-normal, so over the measured range the evidence supports only “unsaturated”. By subsection 5.9 the lognormal value of  $\rho_n$  is already decaying at  $n = 65536$ , so we claim nothing about the asymptotic exponent here; subsection 5.8 and subsection 3.6 give the complete conclusion.

(iii) *The “threshold-like” behaviour of ReLU is a misjudgement.* At  $L_m = 2$  the value of  $\mathcal{I}_n$  for ReLU is close to 1 for  $n \leq 128$  and rises afterwards, so a finite range alone suggests a “threshold effect”; but the local log-slope increases monotonically with  $n$ : -0.030, -0.014, 0.030, 0.130, 0.435, 0.818, **1.195**, which is a transition from below the detection limit to superlinear growth, not a threshold tending to zero.

*Non-monotonicity of the mantissa-width dependence.* The lognormal has a tail slope of 0.833 at  $L_m = 2$  and 1.134 at  $L_m = 4$ , whereas ReLU and the half-normal behave in the opposite way, with almost no effect at  $L_m = 4$  and a marked one at  $L_m = 2$ . This follows from the dead-zone law  $\rho \propto 4^{-L_m}$  ((19)): each extra mantissa bit shrinks  $\mathcal{I}_n - 1$  by a factor of 4, and the data give ratios of 16.3 (ReLU) and 16.9 (half-normal), consistent with  $4^2 = 16$  for two mantissa bits. The lognormal value of  $\rho$  is nearly the same at the two widths (ratio 1.12) because its step lies near the peak of  $|\delta(s)|$ , where the scaling law does not apply; a rigorous characterization of  $\delta$  in that transition region remains open.

**5.4. Sensitivity to the rounding rule.** To rule out that the conclusions are merely an artefact of round-to-nearest, we compared four rounding rules at  $n = 32$  and  $L_m = 4$ : round-to-nearest, round-half-away-from-zero, truncation toward zero, and stochastic rounding

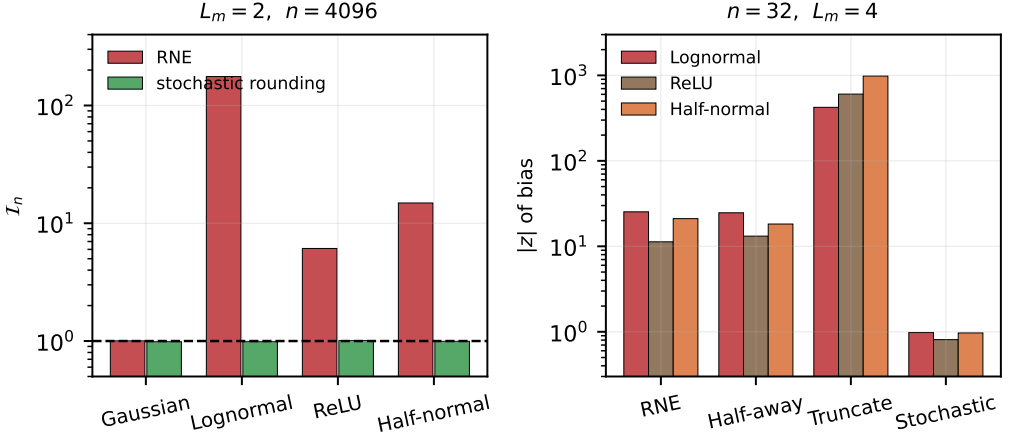


FIG. 3. A falsifiable test of the mechanism, and the rounding-rule control. Left: comparison of  $\mathcal{I}_n$  under round-to-nearest and stochastic rounding at  $n = 4096$ ,  $L_m = 2$  (logarithmic vertical axis; the black dashed line is the predicted value 1). The same distributions, the same block lengths and the same code path, with only the rounding rule changed: the lognormal drops from 176.05 to 0.988. Right: significance  $|z|$  of the block-mean bias under four rounding rules ( $n = 32$ ,  $L_m = 4$ , logarithmic axis). Round-half-away gives almost the same bias as round-to-nearest, truncation amplifies it sharply, and stochastic rounding removes it completely.

(Table 5). Two conclusions follow, and both are visible in Figure 3 (right). First, the finding of subsection 5.1 is not an artefact of round-to-nearest: round-half-away gives almost the same bias,  $-25.33$  versus  $-24.68$  for the lognormal (and  $-11.29$  versus  $-13.16$  for ReLU). Second, stochastic rounding removes the bias completely ( $-25.33 \rightarrow -0.98$  for the lognormal,  $-11.29 \rightarrow -0.81$  for ReLU,  $-21.08 \rightarrow 0.97$  for the half-normal), consistent with Proposition 5, while truncation amplifies it sharply ( $-422.71$ ,  $-603.77$  and  $-982.14$  respectively), in agreement with the engineering consensus that truncation is unusable. The same runs give a self-check on the variance ratio: under SR it is 2.000 for the Gaussian and 2.002 for the half-normal, the value corresponding to the theoretical SR error variance  $s^2/6 = 2 \cdot s^2/12$ , which shows that the  $s^2/12$  of the classical formula is specific to round-to-nearest. Their  $\mathcal{I}_{32}$  values under SR (0.825–1.079) deviate noticeably from 1 because a single group of 1000 blocks is used at  $n = 32$ ; the definitive test of Proposition 5 uses  $20 \times 1000$  blocks and gives  $24/24$  configurations within  $1.000 \pm 2.3\text{se}$  for  $n \geq 512$  (Table 6).

**5.5. A falsifiable test of the mechanism.** Proposition 5 yields a high-risk inference about the mechanism: after switching to stochastic rounding,  $\mathcal{I}_n$  must equal 1 exactly, independently of the distribution, the block length and the mantissa width; any significant deviation would refute Proposition 2 or Proposition 12.

**Result.** All 24 stochastic rounding configurations fall within  $1.000 \pm 3\text{se}$  (0.974–1.061). Table 6 gives the values at  $n = 4096$ ; at  $n = 32$  and  $n = 512$  the SR values are, at  $L_m = 2$ , 1.013, 0.983, 1.004, 1.013 and 1.003, 1.013, 0.996, 1.000 (Gaussian, lognormal, ReLU, half-normal), and at  $L_m = 4$ , 0.992, 1.061, 1.005, 1.004 and 1.010, 1.036, 1.003, 0.974. The corresponding RNE values are 1.010, 3.134, 1.012, 1.007 and 0.999, 31.507, 1.110, 1.473 at  $L_m = 2$ , and 1.015, 1.342, 1.012, 1.023 and 1.009, 15.798, 0.996, 1.039 at  $L_m = 4$ . The contrast is unmistakable: changing only the rounding rule drops  $\mathcal{I}_n$  at  $n = 4096$  from 176 to 0.99.

**5.6. Direct decomposition conditioned on  $\xi$ , and a test of assumption (A2).** The identity of Proposition 2 splits the covariance into two terms, and (A2) asserts that the second is everything. We measure both directly, as in subsection 5.3 (20 groups  $\times$  1000 blocks): bins are the integer values of  $\xi$ , centred by the *within-bin mean*  $m_b$  rather than the within-block

TABLE 6

Test of the mechanism at  $n = 4096$ : is  $\mathcal{I}_n$  identically 1 under stochastic rounding (the predicted value)? Standard errors are in parentheses;  $\Delta$  is the deviation of SR from 1 in standard errors. The same comparison at  $n = 32$  and  $n = 512$  is reported in the text.

| $L_m$ | Distribution | RNE                    | SR                   | $\Delta$ |
|-------|--------------|------------------------|----------------------|----------|
| 2     | Gaussian     | 1.004 (0.009)          | 0.988 (0.012)        | -1.0     |
| 2     | Lognormal    | <b>176.051</b> (1.373) | <b>0.988</b> (0.012) | -1.0     |
| 2     | ReLU         | <b>6.111</b> (0.074)   | <b>1.009</b> (0.012) | 0.8      |
| 2     | Half-normal  | <b>14.871</b> (0.094)  | <b>0.994</b> (0.012) | -0.5     |
| 4     | Gaussian     | 1.009 (0.010)          | 0.975 (0.011)        | -2.3     |
| 4     | Lognormal    | <b>168.274</b> (3.073) | <b>0.991</b> (0.014) | -0.6     |
| 4     | ReLU         | 1.290 (0.014)          | 1.000 (0.010)        | 0.0      |
| 4     | Half-normal  | 1.775 (0.023)          | 0.991 (0.013)        | -0.7     |

TABLE 7

Direct measurement of the conditional decomposition ( $L_m = 2$ ,  $n = 4096$ , round-to-nearest). Cov is the global pairwise covariance;  $\text{Var}(m)$  and  $\mathbb{E}[c]$  are the two terms estimated by binning on  $\xi$ ; the last column is  $\mathbb{E}[c(\xi)]/\text{Cov}$ .

| Distribution   | $\mathcal{I}_{4096}$ | $\text{Cov}(e_i, e_j)$ | $\text{Var}(m(\xi))$   | $\mathbb{E}[c(\xi)]$   | $\mathbb{E}[c]/\text{Cov}$ |
|----------------|----------------------|------------------------|------------------------|------------------------|----------------------------|
| Gaussian       | 0.988                | $-1.02 \times 10^{-7}$ | $-7.3 \times 10^{-10}$ | $-1.02 \times 10^{-7}$ | 0.99                       |
| Student- $t_3$ | 0.997                | $-9.23 \times 10^{-7}$ | $-4.5 \times 10^{-8}$  | $-8.78 \times 10^{-7}$ | 0.95                       |
| Spike mixture  | 0.985                | $-4.30 \times 10^{-9}$ | $-6.5 \times 10^{-11}$ | $-4.24 \times 10^{-9}$ | 0.99                       |
| Lognormal      | <b>175.77</b>        | $6.79 \times 10^{-2}$  | $6.79 \times 10^{-2}$  | $5.5 \times 10^{-9}$   | $8.1 \times 10^{-8}$       |
| ReLU           | <b>6.163</b>         | $1.80 \times 10^{-5}$  | $1.79 \times 10^{-5}$  | $4.5 \times 10^{-8}$   | $2.5 \times 10^{-3}$       |
| Half-normal    | <b>14.880</b>        | $1.19 \times 10^{-4}$  | $1.19 \times 10^{-4}$  | $4.6 \times 10^{-8}$   | $3.9 \times 10^{-4}$       |

mean (see subsection 4.2), keeping only bins with at least 20 blocks (retained mass  $\geq 0.999$ ).

Table 7 reports the measurements; four findings follow.

(1) *Three paths agree; two identities at machine precision.* For each configuration  $\mathcal{I}_n$  is produced in three ways: by (23), by  $1 + (n - 1)\rho$  recovery from the global pairwise covariance, and by  $1 + (n - 1)\text{Var}(m)/\text{Var}(e)$  from binning. The first two agree to four decimal places over all 72 configurations; over the 11 with inflation ( $\mathcal{I}_n > 1.2$ ) the binned and direct estimates differ by at most 5.7% (lognormal,  $L_m = 4$ ,  $n = 512$ ), the remainder within 3%. On the retained bins the relative closure error of  $\text{Var}(e) = \mathbb{E}[\text{Var}(e | \xi)] + \text{Var}(m(\xi))$  is  $3.4 \times 10^{-16}$ , and that of the block-mean decomposition  $\text{Var}(\bar{e}) = \text{Var}(m) + \mathbb{E}[\text{Var}(e | \xi)]/n + (n - 1)\mathbb{E}[c(\xi)]/n$  is  $2.2 \times 10^{-15}$ ; its third term is exactly the contribution of the violation of (A2) and also decomposes the  $z$  statistic of subsection 5.1.

(2) *Share from the violation of (A2); numerical validity of (A2).* The share  $(n - 1)\mathbb{E}[c(\xi)]/\text{Var}(e)$  of  $\mathcal{I}_n - 1$  from  $\mathbb{E}[c(\xi)]$  is at most 1.6 over all 36 round-to-nearest configurations, while the inflated ones reach  $\mathcal{I}_n - 1 = 175$ : it is 1.1% in the worst of the 11 configurations with  $\mathcal{I}_n > 1.2$  and never exceeds 2% under substantial inflation. The ratio  $\mathbb{E}[c(\xi)]/\text{Cov}$  is  $8 \times 10^{-8}$  (lognormal),  $2.5 \times 10^{-3}$  (ReLU) and  $3.9 \times 10^{-4}$  (half-normal); for the three symmetric distributions it is meaningless, Cov already lying below  $10^{-6}$ . Essentially all inflation therefore comes from  $\text{Var}(m(\xi))$ , which is the numerical content of (A2).

(3) *The upper bound holds; a candidate lower bound is refuted.* All 72 configurations satisfy  $0 \leq \mathcal{I}_n \leq n$  (Corollary 3). The candidate lower bound  $\mathcal{I}_n \geq n(\mathbb{E}[e])^2/\text{Var}(e)$  does not hold: for the lognormal at  $L_m = 2$ ,  $n = 4096$  its right-hand side is 1586 while the measured  $\mathcal{I}_n = 175.8$ , and 6 of the 36 round-to-nearest configurations violate it. The bias  $\mathbb{E}[e] = \mathbb{E}[m(\xi)]$  and the within-block correlation  $\text{Cov} = \text{Var}(m(\xi)) + \mathbb{E}[c(\xi)]$  are both moments of the same  $m(\xi)$ , testing only  $m \equiv 0$ ; neither being nonzero implies the other is.

TABLE 8

Parameter-free predictions of the reduced model against measurement (round-to-nearest, 20 groups  $\times$  1000 blocks). Except for the two  $n = 32$  entries of the lognormal, which are given because they are the largest errors at small  $n$ , the table lists the  $n = 4096$  configurations; the model also predicts  $\mathbb{E}[e]$  and  $\text{Var}(e)$  (not listed; maximum relative error  $< 2\%$ ).

| $L_m$ | Distribution          | $\mathcal{I}_n$ measured | $\mathcal{I}_n$ model | Relative error |
|-------|-----------------------|--------------------------|-----------------------|----------------|
| 2     | Gaussian              | 0.9883                   | 1.0003                | +1.2%          |
| 2     | Student- $t_3$        | 0.9965                   | 1.0005                | +0.4%          |
| 2     | Lognormal, $n = 32$   | 3.1354                   | 3.2013                | +2.1%          |
| 2     | Lognormal, $n = 4096$ | 175.7679                 | 176.4593              | +0.4%          |
| 2     | Spike mix.            | 0.9853                   | 1.0000                | +1.5%          |
| 2     | ReLU                  | <b>6.1631</b>            | 6.3155                | +2.5%          |
| 2     | Half-normal           | <b>14.8804</b>           | 15.0719               | +1.3%          |
| 4     | Gaussian              | 1.0014                   | 1.0019                | +0.1%          |
| 4     | Student- $t_3$        | 0.9888                   | 1.0010                | +1.2%          |
| 4     | Lognormal             | 157.1972                 | 156.7585              | -0.3%          |
| 4     | Spike mix.            | 0.9959                   | 1.0000                | +0.4%          |
| 4     | ReLU                  | <b>1.3239</b>            | 1.3646                | +3.1%          |
| 4     | Half-normal           | <b>1.8243</b>            | 1.8868                | +3.4%          |

(4) *Stochastic rounding control.* Under stochastic rounding both terms fall to numerical zero ( $|\text{Var}(m)| \leq 4 \times 10^{-6}$ ,  $|\mathbb{E}[c]| \leq 2.6 \times 10^{-4}$ , every configuration with  $\mathcal{I}_n$  within  $1.000 \pm 0.03$ ), consistent with Proposition 5:  $\mathbb{E}[c(\xi)] = 0$  is not an extra assumption, holding automatically when  $m \equiv 0$ .

**5.7. Parameter-free predictions of the reduced model.** Finally we test the reduced model (17) and (18) of subsection 3.4. Per configuration we estimate the one-dimensional  $\delta(s), \gamma(s)$  from (16) on  $4 \times 10^6$  samples (independently of any blocking), take the empirical distribution of  $s$  (the order statistic of the block maximum) and compute  $\mathbb{E}[e]$ ,  $\text{Var}(e)$  and  $\mathcal{I}_n$  from (18); apart from the sample size there is no adjustable parameter, and  $\delta, \gamma$  are not fitted to the simulation data.

Over all 36 round-to-nearest configurations the relative deviation  $\mathcal{I}_n^{\text{model}}/\mathcal{I}_n^{\text{measured}} - 1$  is at most 4.47% in absolute value (median 1.15%), with  $\mathbb{E}[e]$  and  $\text{Var}(e)$  predicted to below 2%, mostly 0.5%; the  $n = 32$  configurations, at which the model is at its least accurate, are within the same bound. The two-orders-of-magnitude growth in Figure 2 is thus a direct output of (18), not an empirical regularity requiring a separate fit.

**A boundary that must be observed.** The two readings of  $\rho_n$  — the decomposition  $\rho_n = \text{Var}(m(\xi))/\text{Var}(e)$  and the ratio  $\mathbb{E}[s^2\delta(s)^2]/\mathbb{E}[s^2\gamma(s)^2]$  of direct expectations over  $s$  — differ by the same subtraction of  $(\mathbb{E}e)^2$  in numerator and denominator. That is negligible for most configurations here ( $(\mathbb{E}e)^2/\text{Var}(e) \approx 5 \times 10^{-6}$  for the lognormal at  $L_m = 2$ ,  $n = 4096$ ), but for Pareto(4), where  $\mathbb{E}[m] \approx -0.44$  is not small, the two are not interchangeable and one must use the decomposition, as the “model” values here and in Table 10 do.

**Two structural facts.** First,  $\delta$  and  $\gamma$  in (16) depend only on  $s$ , not on  $L_m$ : the lognormal gives the same  $s = 4$  at  $(L_m, \xi) = (2, 4)$  and  $(4, 6)$ , with binned  $\delta = -0.11221$  and  $-0.11223$ , and at  $(2, 5)$  and  $(4, 7)$ , both  $s = 8$ , the values  $-0.11461$  and  $-0.11462$ , supporting the main-regime conclusion of Lemma 8. Second, the dead-zone law is linear: with our sign convention  $\delta = \mathbb{E}[\text{RN}(x/s) - x/s]$ , hence negative,  $\delta(0.125) = -0.00414$ ,  $\delta(0.25) = -0.00841$ ,  $\delta(0.5) = -0.01668$  and  $\delta(1) = -0.03434$ , that is  $\delta(s) = -0.0334 s$ . For ReLU,  $\delta$  is exactly half the half-normal value ( $-0.0167 s$ : a truncated half-normal minus an atom of  $p_0 = 1/2$ , the standard half-normal having  $\delta = -0.0334 s$ ), while  $12\gamma \equiv 1.000$  (half-normal) and  $0.500$  (ReLU) in that range. Hence  $\rho \propto 4^{-L_m}$  in (19).

TABLE 9

Measured  $\mathcal{I}_n$  for Pareto(4),  $L_m = 2$  (independently re-implemented). Rows 1–3 vary  $n$  with 4000/3000/30000/4000 blocks; row 4 holds  $n = 262144$  fixed while the number of blocks grows through  $10^3, 3 \times 10^3, 9 \times 10^3, 2.7 \times 10^4$ . The last row gives two independent measurements at  $n = 10^6$  with 1200 and 30000 blocks.

| $n$                            | 64k                                                         | 256k  | 1M    | 10M   |
|--------------------------------|-------------------------------------------------------------|-------|-------|-------|
| $\mathcal{I}_n$                | 45426                                                       | 41421 | 36065 | 9622  |
| $\mathcal{I}_n/\sqrt{n}$       | 177.4                                                       | 80.9  | 36.1  | 3.04  |
| $\mathcal{I}_n/n$ ( $n$ fixed) | 0.125                                                       | 0.158 | 0.146 | 0.135 |
| $n = 1\text{M}, 1200$ blocks   | $\mathcal{I}_n = 37692$ (4.5% from 36065 with 30000 blocks) |       |       |       |

TABLE 10

Bounded versus unbounded support ( $L_m = 2$ ). The bounded distributions are supported on  $[0, 1]$ ,  $[0, 1]$  and  $[0, 2]$  respectively; all others are unbounded. Values of  $\mathbb{P}(\xi = \text{mode})$  and of the model  $\mathcal{I}_n$  are taken at  $n = 4096$ .

| Distribution          | $n = 32$ | $n = 512$ | $n = 4096$ | $\mathbb{P}(\xi = \text{mode})$ | $\mathcal{I}_n$ model |
|-----------------------|----------|-----------|------------|---------------------------------|-----------------------|
| Beta(2, 5)            | 1.001    | 1.007     | 1.009      | 1.000                           | 1.000                 |
| Beta(5, 2)            | 1.014    | 1.007     | 0.999      | 1.000                           | 1.000                 |
| Truncated half-normal | 0.999    | 0.993     | 0.995      | 1.000                           | 1.000                 |
| Exponential           | 1.235    | 8.284     | 41.885     | 0.747                           | 41.448                |
| Pareto(4)             | 7.154    | 290.22    | 2886.30    | 0.573                           | 2887.29               |
| Lognormal             | 3.197    | 30.537    | 175.19     | 0.602                           | 176.04                |

**5.8. Pushing the block length further.** Beyond  $n \approx 10^5$  for Pareto(4), the measured  $\mathcal{I}_n$  no longer tracks a constant- $\rho_n$  extrapolation. Table 9 records the large- $n$  measurements and the corresponding population prediction of the reduced model.

As  $n$  increases from 64k to  $10^7$  (a factor of  $1.6 \times 10^5$ ),  $\mathcal{I}_n$  decreases from  $4.5 \times 10^4$  to  $9.6 \times 10^3$  and  $\mathcal{I}_n/\sqrt{n}$  from 177 to 3.0 (so  $\mathcal{I}_n/n$  falls from 0.693 to 0.00096). A constant  $\rho_n \approx 0.70$  would predict multiplication by 16 when  $n$  grows by 16, and that behaviour holds only for  $n \leq 65536$ ; comparing  $n = 10^6$  with  $n = 10^7$  gives a local log-slope  $\log(36065/9622)/\log 10 = 0.57$ , consistent with  $\mathcal{I}_n \leq n$  (Corollary 3) and ruling out asymptotically linear growth. Proposition 12 states that  $B_n = \text{Var}(m(\xi)) \rightarrow 0$  and  $\rho_n \rightarrow 0$ , hence  $\mathcal{I}_n = o(n)$ : linear accumulation is impossible for any unbounded distribution with finite second moment. Evaluating (18) exactly by one-dimensional quadrature (see the remark in subsection 3.6) gives  $3.61 \times 10^4$  at  $n = 10^6$  and  $9.2 \times 10^3$  at  $n = 10^7$ , matching the measured  $3.61 \times 10^4$  and  $9.6 \times 10^3$ : the fall-off is the decline of the population curve past its peak, not an artefact of insufficient sampling, and the row with  $n$  fixed shows stability,  $\mathcal{I}_n/n$  staying within 0.125–0.158 as the block count grows by a factor of 27. This paper claims only that the independence premise fails for asymmetric inputs, and claims no particular asymptotic order of accumulation: what is a theorem is  $\mathcal{I}_n \leq n$  (Corollary 3),  $\mathcal{I}_n \rightarrow 1$  for bounded support (Proposition 10) and  $\mathcal{I}_n = o(n)$  for unbounded support (Proposition 12).

**5.9. Numerical comparison of bounded and unbounded support.** This subsection separates the two factors “bounded versus unbounded” and “light versus heavy tails” using five new distributions: Beta(2, 5), Beta(5, 2) and a half-normal truncated to  $[0, 2]$  (bounded, asymmetric), the exponential distribution (unbounded, light-tailed) and Pareto(4) (unbounded, regularly varying tail). The methodology is that of subsection 5.6.

Table 10 fully supports Proposition 10: the three bounded-support distributions give  $\mathcal{I}_n = 1.00 \pm 0.01$  for  $n \leq 4096$ , indistinguishable from symmetric distributions and independently of the endpoint tail index. The unbounded side is the opposite: the exponential distribution already reaches 41.9 at  $n = 4096$  and Pareto(4) reaches 2886, and the prediction error of the

TABLE 11

$\rho_n = (\mathcal{I}_n - 1)/(n - 1)$  at large block lengths ( $L_m = 2, 20$  groups  $\times$  500 blocks =  $10^4$  blocks per configuration, averaged over two independent seeds; the seed-to-seed difference is below 1% for Pareto(4) and the lognormal at all three block lengths, and reaches 5.1% for the half-normal at  $n = 65536$ , where  $\mathcal{I}_n$  is small and the lattice oscillations of subsection 5.8 dominate).

| Distribution | $\rho_{4096}$ | $\rho_{16384}$ | $\rho_{65536}$ | $\mathcal{I}_{65536}/\mathcal{I}_{4096}$ |
|--------------|---------------|----------------|----------------|------------------------------------------|
| Pareto(4)    | 0.7038        | 0.7423         | 0.6958         | <b>15.77</b>                             |
| Lognormal    | 0.0427        | 0.0181         | 0.0177         | 6.56                                     |
| Half-normal  | 0.0034        | 0.0025         | 0.0001         | 0.60                                     |

TABLE 12

*Explanatory power of the mechanism for the individual observations.*

| Observation                                                                                                                              | Mechanistic explanation                                                                                                                                                               | Status                                          |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Symmetric distributions give exactly $\mathcal{I}_n = 1.0$ for $n \leq 4096$                                                             | $F$ symmetric $\Rightarrow \delta(s) \equiv 0$ , independently of (A2)                                                                                                                | verified (exact)                                |
| The within-block correlation comes from $\text{Var}(m(\xi))$ , not from $\mathbb{E}[c(\xi)]$ ; the asymptotic exponent does not exceed 1 | exact decomposition of Proposition 2, with the violation of (A2) accounting for at most 2% of $\mathcal{I}_n - 1$ ; and Corollary 3 (Cauchy-Schwarz)                                  | verified (direct measurement);<br>proved        |
| The “threshold-like onset” of ReLU, and its variance ratio 0.506                                                                         | dead-zone law $\rho \propto 4^{-L_m}$ ((19)), measured ratio 16.3 against $4^2 = 16$ ; atom $p_0 = 1/2$ gives $\gamma = 1/24$ and ratio $1 - p_0$ , and $2(1 - p_0) = 1.000$ under SR | explained quantitatively                        |
| All numerical values of $\mathcal{I}_n$                                                                                                  | (18), maximum error 4.5% over 36 configurations (no fitted parameter)                                                                                                                 | verified                                        |
| $\mathcal{I}_n \equiv 1$ under stochastic rounding                                                                                       | Proposition 5 ( $m \equiv 0$ and $c \equiv 0$ )                                                                                                                                       | verified (24/24)                                |
| Bounded support reaches a plateau; unbounded support rises and then falls                                                                | bounded: $\xi$ concentrates on one grid point $\Rightarrow B_n = O(p^n)$ ; unbounded: $m(\xi) \rightarrow -\mathbb{E}[x_1] \Rightarrow B_n \rightarrow 0$ (Propositions 10 and 12)    | proved; the peak location is given by the model |

reduced model is below 2% for all six distributions, both widths and all three block lengths. By Proposition 12 the inflation on the unbounded side must eventually fall back: the peak of Pareto(4),  $5.9 \times 10^4$ , lies at  $n \approx 10^5$  (see the remark in subsection 3.6), and the measurements of subsection 5.8 agree with that curve.

Table 11 shows two things. First, for  $n \leq 65536$  the value of  $\rho_n$  for Pareto(4) is roughly constant (0.696–0.742), and a 16-fold increase in  $n$  multiplies  $\mathcal{I}_n$  by 15.77, close to the factor 16.00 predicted by a constant  $\rho_n$ , i.e.  $\text{Var}[\sum_i e_i] \approx 0.70 n^2 \text{Var}(e)$ . This is the basis of the constant- $\rho_n$  extrapolation that “ $\sqrt{n}$  degenerates to  $n$ ” over this range; subsection 5.8 shows that it is merely a *transient* over attainable block lengths. Second, the lognormal value of  $\rho_n$  falls from 0.043 to 0.018, making  $\mathcal{I}_n$  markedly *sublinear* (its  $\mathcal{I}_n/n$  at  $n = 65536$  is 0.0177, against 0.0361 at  $n = 10^6$  in Table 9), while the half-normal exhibits an *absolute* decrease of  $\mathcal{I}_n$  at  $n = 65536$  ( $42.4 \rightarrow 9.0$ ) caused by the phase of  $\log_2 M_n$  relative to the integer grid, a jump the reduced model also reproduces. Under unbounded support  $\mathcal{I}_n$  therefore carries oscillations caused by lattice alignment, and treating it as a smooth power law is dangerous.

**6. Discussion.** Proposition 2, Proposition 7 and the reduced model (17) give one picture: the within-block errors are coupled through the shared step  $s$ , the rounded logarithmic scale of the largest magnitude in the block, so every fluctuation of  $s$  enters *every* error with the same sign, with coefficient the deterministic response curve  $\delta(s)$ . If  $\delta \not\equiv 0$ , that common fluctuation accumulates coherently in a summation instead of averaging away; and since  $\delta, \gamma$  are one-dimensional, all the randomness resides in the distribution of  $s$ . Table 12 collects the mechanism’s explanatory power for the observations of section 5; subsection 6.2 states the boundary of validity.

**6.1. Scope of validity and practical implications.** The effect varies in strength across distributions: at the MX block length  $n = 32$  the lognormal gives  $\mathcal{I}_{32} = 3.08$ , a summation error 1.77 times the independent-model prediction, whereas ReLU outputs give only 1.037 (about 1.8%). This paper therefore cannot be read as claiming that current hardware is already seriously inaccurate. By subsection 5.9 the  $\sqrt{n}$  advantage is markedly weakened only for unbounded support with a *regularly varying tail* (Pareto type) and only for  $n \lesssim 10^5$ ; bounded support is better behaved than the classical model, and fast-decaying unbounded tails such as the lognormal lie between the two. Since Proposition 12 forces the unbounded-side inflation to fall back for large  $n$ , the value of the classification is in showing that block length and tail type jointly determine the risk, not in giving an extrapolatable power law. Finally, Proposition 5 concerns the accumulation statistics of the within-block error only: for asymmetric activations stochastic rounding removes the bias and *exactly* restores  $\sqrt{n}$  accumulation (subsection 5.5, 24/24 configurations). This is separate from end-to-end training stability: Cim et al. [7] find that on native FP4 hardware, once the weight gradient uses MXFP4, *neither* stochastic rounding nor random Hadamard rotations stabilize training, whereas a deterministic Hadamard transform does, the instability arising from “structured micro-scaling errors along sensitive gradient paths, rather than insufficient stochasticity”.

## 6.2. Limitations.

1. The rate of the asymptotic decay is undetermined: rigorous are only  $\mathcal{I}_n \leq n$  (Corollary 3),  $\mathcal{I}_n \rightarrow 1$  for bounded support (Proposition 10) and  $\mathcal{I}_n = o(n)$  for unbounded support (Proposition 12). Proposition 12 gives no rate, so  $\mathcal{I}_n \rightarrow 1$  rests on the numerical model (see subsection 3.6) and the peak location is not a theorem; a rate theorem would need a regularity assumption on  $\delta(s)$  as  $s \rightarrow \infty$  that is independent of  $n$ .
2. The reduced model is the leading asymptotic term of the main regime. Propositions 2 to 7, Lemma 8 and Corollaries 3 to 6 are rigorous, while (17) discards the  $O(\varepsilon)$  term of Lemma 8, whose inequality degenerates when  $2^\xi$  lies far below the data scale (Lemma 8). It matches the simulations to a few per cent in the experimental main regime but is untested in the dead-zone-dominated one, and the  $L_m$  scaling law  $\rho \propto 4^{-L_m}$  ((19)) likewise fails near the peak of  $|\delta(s)|$ : the transition region lacks a rigorous description, and the test of (A2) in subsection 5.6 is binned, hence numerical rather than analytic.
3. The format model is idealized, and two observations are informal. Equation (1) assumes the shared exponent to be the rounded logarithmic scale of the largest magnitude in the block, round-to-nearest rounding, and no overflow; we tested four rounding rules but not real hardware, and accumulator width, sticky-bit handling and normalization in tensor cores could all change the conclusions. Separately, in initial experiments the growth order of the dot-product error under a shared exponent was worse than under an elementwise exponent, but no same-standard comparison was carried out, and the literature search comprised two focused passes.

**7. Conclusions.** This paper re-examines a statistical assumption carried in block floating point error analyses since 1996: that within-block quantization errors are zero-mean and uncorrelated. The covariance identity  $\text{Cov}(e_i, e_j) = \mathbb{E}[\text{Cov}(e_i, e_j \mid \xi)] + \text{Var}(m(\xi))$  needs no extra assumptions, conditional exchangeability being automatic, and Cauchy–Schwarz gives  $0 \leq \mathcal{I}_n \leq n$ , so any asymptotic log–log slope is at most 1; Proposition 12 sharpens this to  $\mathcal{I}_n = o(n)$  for unbounded support with  $\mathbb{E}[x_i^2] < \infty$ . The variance law separates two independent defects of the classical model:  $\delta \neq 0$  creates correlation, and  $\gamma \neq 1/12$  biases the variance formula. Under stochastic rounding both terms vanish, so  $\mathcal{I}_n \equiv 1$ .

The numerical results support the same picture. For symmetric distributions, (H1) and (H3) hold in the measured range; for asymmetric ones, a direct decomposition given  $\xi$  shows that the

within-block correlation comes almost entirely from  $\text{Var}(m(\xi))$ . With no fitted parameter, the reduced model reproduces measured  $\mathcal{I}_n$  within 4.5% over 36 round-to-nearest configurations, predicts the dead-zone scaling  $\rho \propto 4^{-L_m}$  and the ReLU variance ratio 0.506, and yields 1.000 under stochastic rounding by the same atom argument.

The asymptotic classification is clear: bounded support forces  $\mathcal{I}_n \rightarrow 1$  at an exponential rate (Proposition 10), while unbounded support produces an inflation that first rises and then falls, Proposition 12 ruling out asymptotic linear accumulation for any distribution in this class with finite second moment. Large- $n$  measurements and exact quadrature of the reduced model agree on this fall-off (subsection 5.8).

The practical message is narrow but concrete: current hardware need not already be inaccurate at the MX block length  $n = 32$ , the effect size depending strongly on the input distribution. What does fail, once the data are asymmetric, is the independence premise used to justify  $\sqrt{n}$  accumulation in probabilistic error analyses. BFP models applied to neural-network activations should therefore be corrected, and the scale of that failure is controlled: it appears at moderate block lengths and then decays, rather than growing without bound.

## Appendix A. Data availability.

The complete tables behind section 5 — the full grid of  $\mathcal{I}_n$  over six distributions, two mantissa widths and seven block lengths, and the stochastic-rounding comparison at  $n = 32, 512$  and  $4096$  — are distributed with the reproduction package described in section 4, together with the archived terminal output of every batch of runs. The tables printed in section 5 are the subsets that the argument in this paper uses; the only values quoted in the text but not tabulated there are the  $n = 32$  and  $n = 512$  entries of Table 6 and the  $z$  values of Figure 1 (top), all of which appear in the package.

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